Indian Institute of Information Technology Vadodara

Embedded System for Nutrient Stress Detection

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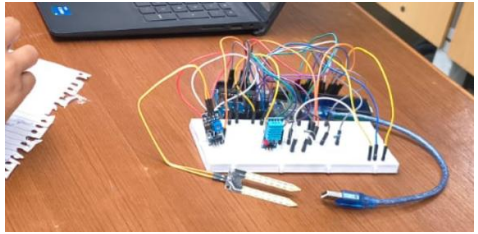
Course Name: Embedded System Project



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Introduction

Agriculture plays a crucial role in human life
Plant growth depends on proper nutrient availability
Nutrient stress reduces crop yield and quality

Traditional detection methods:

- Time consuming
- Expensive
- Require laboratory testing

Conclusion :
Need for low-cost, real-time smart monitoring system

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Research Paper

Paper: Smart Agriculture for Sustainable Growth

- https://dl.acm.org/doi/10.1145/3647444.3647922

Uses IoT sensors to collect soil data:
NPK values
pH
Temperature
Data is analyzed using AI models

Key Idea:
Real-time monitoring improves decision making in agriculture

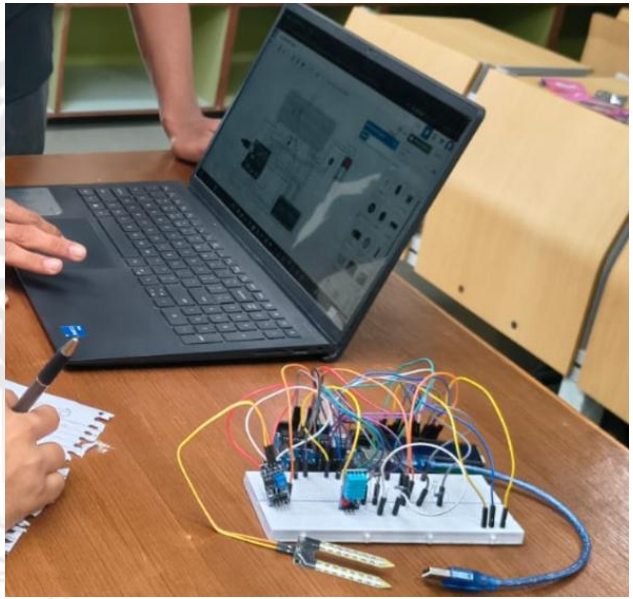
Research Paper

Literature Review (Model)

Content:
Uses Parallel Hybrid CNN-LSTM model
CNN:
Extracts spatial relationships
LSTM:
Captures temporal patterns

Bayesian Optimization improves performance

Result:
Achieved high accuracy (~98%)



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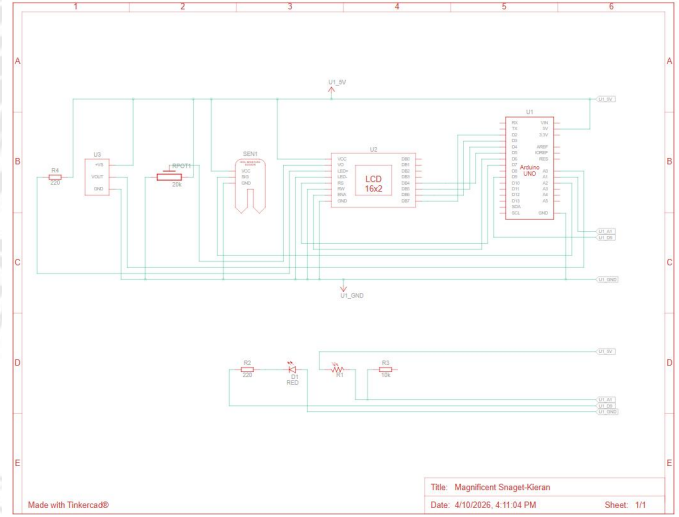
Objectives and Comparison

Content:

Objectives:
Detect nutrient stress in plants
Develop low-cost IoT system
Implement ML model for prediction

What Research Paper Did:
IoT + AI + cloud-based system
Uses advanced sensors

What I Did:
Arduino-based IoT system
Threshold-based detection
CNN-LSTM ML model (separately implemented)



Schematic View of Simulation

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Methodology

Content:

System Flow:

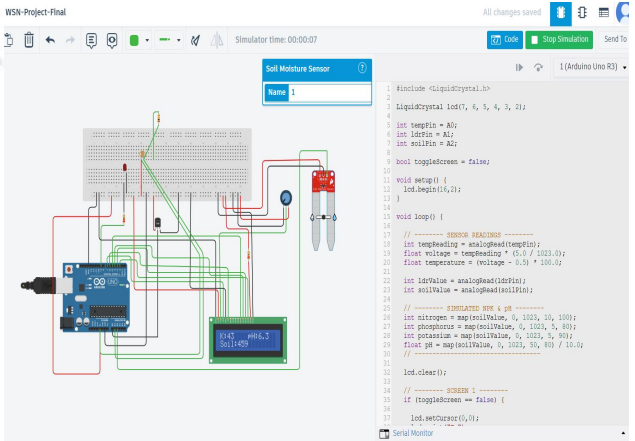
Sensors → Arduino → Processing → ML Model → Output

Steps:

- Sensor data collection + Government Soil Datacard- used for more Data
- Threshold-based stress detection
- Dataset preprocessing (N, P, K, pH)
- CNN-LSTM model training
- Prediction of soil condition

Live Simulation Link Tinkercad

https://www.tinkercad.com/things/iZoJfhCpb4h-embedded-project/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard%2Fdesigns%2Fcircuits



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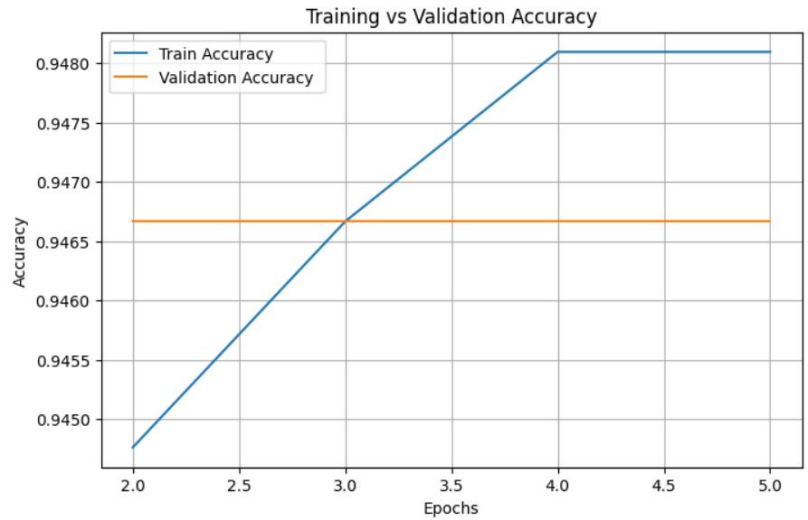
RESULTS

Content:
Shows training vs validation accuracy
Model learns effectively over epochs

Overfitting reduced using:
Dropout
Early stopping
Class weights



Model Accuracy Graph

RESULTS

Model Loss Graph

Content:

Shows training vs validation loss
Loss decreases as model learns

Validation loss stabilizes → good generalization

Model Loss Graph

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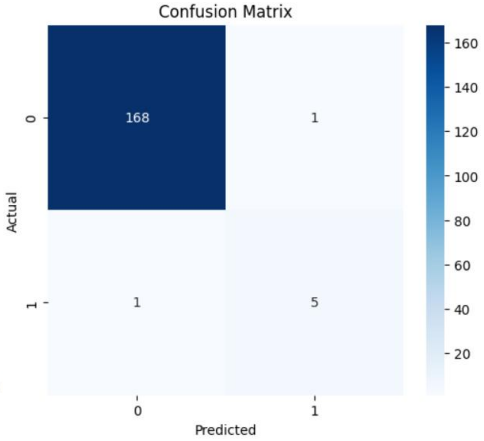
RESULTS

CONFUSION + PREDICTION

Confusion Matrix:
Shows correct and incorrect predictions
Sample Prediction:
Input:
N = 80, P = 40, K = 45, pH = 6.5

Output:
Predicted Condition: Deficient

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1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 165ms/step
Input: [80.  40.  45.   6.5]
Predicted Condition: Normal
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Input: [20. 10. 15.  5.]
Predicted Condition: Deficient
--------------
Input: [120.   90.   95.    8.5]
Predicted Condition: Normal
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```

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REFERENCES

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